Technology Stack

SMARTBRIDGE - Technology Stack

Date: 21/09/2026

Team ID: SWTID-2026-4223

Team Size: 4

Team Leader: Sweatha P

Team Members:Lakshmi Narayanan R, Rithika P, Sandhiya K

Project Name: AI FitTrack

Maximum Marks: 2 Marks

Technology Stack Details

1. Frontend / Client-Side — Technology: Postman / Thunder Client / React.js. Purpose: Provides a flexible client interface to execute HTTP REST API calls and render workout JSON payloads.

2. Backend / Server-Side — Technology: Node.js & Express.js. Purpose: Lightweight, non-blocking asynchronous event loop allowing fast execution of API endpoints and middleware.

3. Database / Data Storage — Technology: MongoDB Atlas & Mongoose ODM. Purpose: Flexible NoSQL document database ideal for dynamic workout schemas and high performance data queries.

4. Cloud / Hosting / Deployment — Technology: MongoDB Atlas Cloud & Local Node Execution. Purpose: Provides auto-scaling cloud database hosting with IP whitelisting and reliable connection URIs.

5. Version Control & CI/CD — Technology: Git & GitHub. Purpose: Enables seamless team collaboration, code history tracking, and repository management.

6. Third-Party APIs / Other Tools — Technology: Google Gemini API (@google/genai). Purpose: Enables AI workout plan generation (gemini-2.5-flash) and personalized fitness insight analytics.